Studying Long-Term Changes in the Economy and Society Using the HISCO Family of Occupational Measures

Article · January 2022

DOI: 10.1093/acrefore/9780190625979.013.541

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Subject: Economic Development , Economic History Online Publication Date: May 2020
DOI: 10.1093/acrefore/9780190625979.013.541

Summary and Keywords

Occupations are a key characteristic for analyzing momentous changes in economy and society. Classical economists rooted their analyses in occupational divisions, emphasizing the division of work and its continuous evolution. Modern economists and economic historians also debate the wealth of nations by looking at the global changes in the labor force, at changing labor force participation rates, at winners and losers in the class structure, and in variations in this across the globe—stressing the importance of human capital for work and of changes therein for economic growth. To study such momentous changes over past centuries, historical occupational data are needed as well as measures and procedures to work with these data systematically and comparatively. The Historical International Standard Classification of Occupations (HISCO) maps occupational titles into a common coding scheme across the globe. HISCO-based measures of economic sector and economic specialization have been derived. To answer a number of interesting questions, the HISCO family has been extended to include HISCO-based measures of social status (HISCAM) and social classes (HISCLASS). Armed with his toolbox, scholars are able to study the development of the economy and society over past centuries.

Keywords: economic growth, economic specialization, industrialization, occupations, labor force participation, HISCAM, HISCLASS

Introduction: Studying Long-Term Changes in Inequality Using Occupations

Inequality has been on the research agenda since at least the days of Marx. The core issues of interest are: How much inequality is there, and how does it vary among countries and across time? How difficult is it to transcend barriers between social layers? What drives inequality and mobility? And what legitimizes it—especially in an age of stagnating mobility? To answer these questions, economists tend to capture inequality in monetary terms, as a product of wealth and income, while sociologists and economic historians tend to use occupations. They argue that occupations are key to analyzing the economy and society; with just modest exaggeration, one might say they are the DNA of the economy and society. The question “What kind of work do you do?” is often one of the first things people ask one another. And with good reason. Work generates income, invests one with social status, and, hopefully, creates personal fulfillment. It is indicative of past education and of present, as well as future, earnings capacity.

Practically speaking, occupations are often the only indicator of social and economic standing in historical data. That is why many economists and historians use occupations, regardless of whether they subscribe to any explicit social theory. And the ubiquity of occupations in the historical record is a clear indication that contemporaries thought them important. While occupational titles have been extensively put to good use for contemporary societies by eminent census officials and scholars of social stratification, until the creation of the Historical International Standard Classification of Occupations (HISCO) this was not the case for past societies. HISCO and its progeny offer scholars a handle to study important long-term changes in economy and society. HISCO is of interest when making comparisons over time and space in any or all specific occupations, occupational groups, or labor force participation rates. HISCO also supports studying processes of economic specialization in the wake of the industrial and other work-related revolutions. The measures derived from HISCO offer useful perspectives on social class and social strata.

In their various economic trajectories, societies in different parts of the world might differ in the organization of work through occupations. The range of occupations open to individuals might differ, as might the distribution across occupational categories. Sorting according to occupation is almost never devoid of issues of gender, age, and ethnicity. Even if occupational sorting is ever free of discrimination in a single country, there would still be marked contrasts between various parts of the world. Many of the more highly specialized and profitable types of work are still more prominent in the Global North than elsewhere. In fact, many of our modern, specialized occupations arose during the economic transformations denoted by the term Industrial Revolution. There have been transformational changes over the course of time in how economies have been organized in terms of work.

It is thus understandable why economists and sociologists have been interested in studying patterns, causes, and consequences of occupational divisions over time and space. Classical economists like Adam Smith emphasized the division of work, its evolution, and the transformational changes it brought to their societies. So, too, do scholars of modern-day labor economics. In their surveys, contemporary sociologists use occupational titles as a staple. Contemporary economists arguably rely more on income and wealth as an indicator of social position, and not without reason. One can input dollars, yuan, yen, euros, etc., to a regression model, but not words, and that is how occupational information comes to us, through occupational titles.

This contribution begins by looking at the diversity of occupational titles and the ways to map these into common coding schemes across the globe, past and present. It pays particular attention to the International Standard Classification of Occupations (ISCO) schemes developed over more than half a century by the International Labor Organization (ILO) in close consultation with representatives of national statistical agencies across the world. ISCO describes contemporary societies. However, it has a historical cousin: HISCO. HISCO is rooted in the 1968 version of ISCO and was developed by economic and social historians to cover occupational titles found in historical censuses and other sources in various parts of the world. It has been widely used to describe occupational variation in the 18th, 19th, and 20th centuries and, to a lesser extent, in earlier periods. On its own, HISCO is of rather more limited use, but it comes as part of a family of taxonomies and procedures, increasing its utility to economists, social scientists, historians, and others interested in occupational analysis. After discussing HISCO, this article proceeds to discuss the family of which it is part. It will discuss HISCO-based measures of economic sector, economic specialization and modernization, and educational skills—illustrated by a few concrete examples from different regions and time periods. Economists will easily be able to recognize them.

However varied the thinking of sociologists about the social stratification of societies, past and present, in this or that part of the world, they are familiar both with the concepts of social class and social status and, in one way or another, with rooting the operationalization of these concepts in workable measures based on occupational titles. The HISCO family of taxonomies might be said to rather unimaginatively borrow—lock, stock, and barrel—from sociological concepts of class and status, but they root them in HISCO, thus allowing one to analyze societal processes not just for contemporary societies but also, in principle, for all societies over past centuries. This article will discuss the concepts of social class and social status, and some of the debates surrounding these, as a basis for discussing HISCO-based measures of social status (HISCAM), large social classes (HISCLASS), and smaller classes (a historical microclass scheme).

The Historical International Standard Classification of Occupations

Occupations—nearly every adult, past or present, has one (although they have not always been noted for women). For centuries, in large parts of the globe, church officials and civil servants have noted these at major life events, from cradle to grave. In many countries, for the past two centuries occupations have been systematically recorded in censuses. Occupations allow for comparisons of work between individuals, communities, countries, and time periods and are a key variable in the social sciences, the life sciences, and the humanities.

The purpose of HISCO is to create an occupational classification system that allows comparisons to be made both internationally and historically (van Leeuwen, Maas, & Miles, 2004). It is closely connected to 21st-century classifications, which allows comparisons up to the present. HISCO originated in ISCO, an occupational classification system developed by the ILO. ILO produced several versions of ISCO in 1958, 1968, 1988, and 2008. HISCO is based on the 1968 version (ISCO68). This version contains a larger number of occupations than the 1958 version, while also including many occupations of a historical nature no longer included in the 1988 version.

HISCO distinguishes 1,675 different occupational categories and describes their tasks and duties. As in ISCO68, each category is characterized by a five-digit numeral code. The last three digits refer to the "unit groups"—298 in all. The first two digits refer to the "minor groups"—76 in all; the first digit refers to the "major groups"—10 in all. For example, codes 6-xx.xx refer to the primary sector of the economy, with codes 6-2x.xx identifying different types of agricultural and animal husbandry workers. The latter includes the code 6-22.xx for field crop and vegetable farm workers. These, in turn, relate to various more specific occupational categories: field crop farm workers (6-22.10), vegetable farm workers (6-22.20), wheat farm workers (6-22.30), cotton farm workers (6-22.40), rice farm workers (6-22.50), and sugar cane farm workers (6-22.60).

The main groups are presented in Table 1. The 1,675 occupational categories were first used to classify data from Belgium, Canada, France, Germany, the Netherlands, Norway, Sweden, and the United Kingdom, and the results were published in the HISCO manual. Later, data from many other countries were classified in HISCO. At present, HISCO is used by many scholars and virtually all leading databases across the world. The temporal coverage is from the 16th to the 20th century; the geographical coverage includes Algeria, Belgium, Brazil, Canada, Denmark, Egypt, England, Finland, France, Germany, Greece, Italy, the Netherlands, Norway, the Philippines, Portugal, Russia, Spain, Sweden, Switzerland, Uganda, Uruguay, the United States, and other regions (see also History of Work website).

Table 1. Major HISCO Groups

Group No.	Label
0/1	Professional, technical, and related workers
2	Administrative and managerial workers
3	Clerical and related workers
4	Sales workers
5	Service workers
6	Agricultural, animal husbandry, and forestry workers; fishermen and hunters
7/8/9	Production and related workers, transport equipment operators, and laborers

Source: van Leeuwen, Maas, and Miles (2004).

Sometimes additional information in the occupational titles is not captured in the HISCO categories. To preserve such information, three auxiliary variables were created: Status (e.g., for noble titles but also for master artisans), Relation (e.g., for retired or wife of), and Product (when the source mentions a product, such as tin or copper, while the HISCO description does not). While Product has not found universal use, Relation and Status are commonly used and are indeed needed for many of the measures and procedures mentioned here. The fact that data covering a long period and from many parts of the globe have been coded using a common coding scheme allows for comparative research on occupational structures, social class formation and its consequences, as well as on processes of status attainment.

There are also drawbacks to using occupational titles from historical sources worldwide. First, the variety of occupational titles is dazzling; there are literally hundreds of thousands, albeit sometimes varying only in spelling or level of detail. The start-up costs of coding all these occupational titles consistently into a scheme have thus been considerable. This problem has become less onerous, however, because of the existence of datasets of coded occupations and a variety of fuzzy coding procedures. Second, like all historical sources, those containing occupations are not perfect. Censuses or marriage records rarely contain occupations such as burglar or sex worker. While illicit or low-status occupations are underrecorded, the variety of self-declared occupations in vital registers is dazzling. This is certainly so in comparison with occupations that were not self-declared but “filtered” by a census taker—the census taker might have been instructed not to include part-time work. Third, in many historical sources there is less information on, and a much smaller range of, occupations for women than for men, reflecting in part the situation of past labor markets and in part under registration. Fourth, at times, and depending on the source, the occupational information is vaguer than in contemporary surveys—although the reverse can also be the case. The fact that production and sales are not always distinguished (reflecting historical realities) can be problematic; petty traders or farm laborers might not always be sufficiently distinguishable from larger traders or farmers. A special subset of this problem relates to “workers” or “laborers” without further information. The vagueness of titles is less problematic for censuses or labor counts, where sometimes additional information has been stored, for example on employment status or size of establishment.

Changes in the Labor Force: Economic Diversification and Changing Labor Force Participation Rates

Once there are comparable data on the occupational activities of individuals across time and space, coded in a common scheme, historical economists and economic historians can study issues of core concern to them. This article briefly discusses the process of economic diversification whereby all countries transitioned from agriculture, as the main economic sector, to industry and later to services, a process that has generally greatly increased national and regional GDP but also played out differently over the world. Subsequently, a related issue is discussed—how to study changes in the labor force participation (LFP) rate using the HISCO family of measures.

Economic Diversification as Measured Using Occupational Diversity

Classical economists and sociologists, arguably more than modern ones, rooted their analyses in the study of changing occupational divisions. To explain the economy and society, they emphasized the division of work and its continuous evolution. When Adam Smith asked himself what caused the new wealth of Great Britain, he first eliminated potential explanations such as gold from the colonies or farmers working harder because ministers made them more religious. He settled on a rise in productivity due to the division of labor, notably in manufacturing, as its true cause. The opening sentence of his opus magnum already emphasizes the division of labor. To explain the wealth of nations, he discussed most famously pin makers. The pin maker, of course, featured prominently in the *Encyclopédie* by Diderot and d’Alembert, a masterpiece intended to summarize all human knowledge. Smith used the pin maker as an example of the evolution of the division of tasks over time and the enormous consequences for the wealth of nations.

A century before Adam Smith, the notion of individual self-interest of occupational bearers had been adduced by Mandeville in his *Fable of the Bees*. He compared the human world to a beehive and stressed that it is the pursuit of individual occupational activities

—including illicit ones such as thievery—to individual advantage that keeps the world turning. Smith was thus not alone in emphasizing occupational divisions as the motor of economic and social change, as is also evident from the writings of Durkheim, Saint-Simon, Marx, and others. Their appreciation of the consequences varied. Smith shared some concerns regarding deskilling and a resulting power shift to employers and alienation due to monotonous labor, but he thought these negative aspects were marginal, could be resolved, and were more than counterbalanced by important other social and economic processes: important changes in the labor force composition, the rise of new and complex occupations, social class distribution, and the growth in GDP.

If the growth in the wealth of nations is due to each nation specializing in what it can produce comparatively cheaply and trading that for the best another nation has to offer, one should see these changes in the production process reflected in occupational change. While some occupational tasks will no longer be needed and will disappear, on the whole the direction is one of occupational specialization. General tasks that would normally be fulfilled by one person will make way for specific, specialized tasks requiring different skills or different machines and tools that will be used by specific individuals. The process of specialization in the production process is thus linked to the process of occupational specialization. Using an occupational classification scheme such as HISCO, one can study this process in two ways: by looking at increasing occupational diversity and by looking at the rise of new industrial (and later service) occupations. There are other ways, too, not discussed here: one could, for instance, look at the ratio between general occupations (generally ending in 00 codes) and more specialized ones. or one could look at dual occupational titles such as farmer-fisherman, which indicate a need to earn an income from various sources.

As indicated, the basic five-digit hiscodes were developed such that all known historical occupational activities, both new and old, could be accommodated. Thus HISCO has, for instance, a box for a Nuclear Physicist, which is of no use for the 18th century, as well as a box for Linotype Operator, which is of no use in the 21st century. At any one moment there are thus more boxes than needed to describe a particular historical society. Generally, more new occupations emerge over time than old ones vanish (especially since a census will still record the sole surviving worker in an almost extinct occupation). Increasing occupational diversity can be witnessed by looking at what proportion of the 1,675 basic hiscodes in an area is covered by at least one person and then looking at this proportion in a later year. Figure 1 shows the occupational diversity of bridegrooms from samples from a set of European countries in the period 1800–1910 (for further details, see Maas & van Leeuwen, 2016). Figure 2 shows the same diversity over a much longer period (1720–1986) for a single country for which there happens to be such data—France (van Leeuwen, Maas, Rébaudo, & Pélissier, 2016). The figures indicate a general pattern of economic specialization in Europe from at least the beginning of the 19th century up to the present, covering both the transition from agriculture to industry and then to service-dominated economies. They also show minor deviations from this general pattern—the number of occupational characteristics in the department of Pas de Calais actually goes down, for example—which might be due to the special characteristics of the datasets or to time- and place-specific developments.

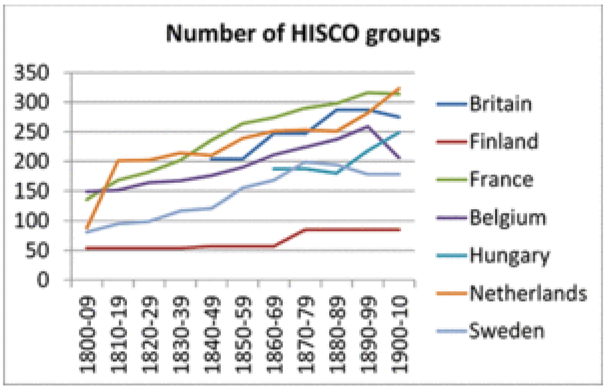

Figure 1. Occupational diversification in European countries, 1800–1910.

Source: author's calculations on same datasets as reported in Maas & van Leeuwen, 2016.

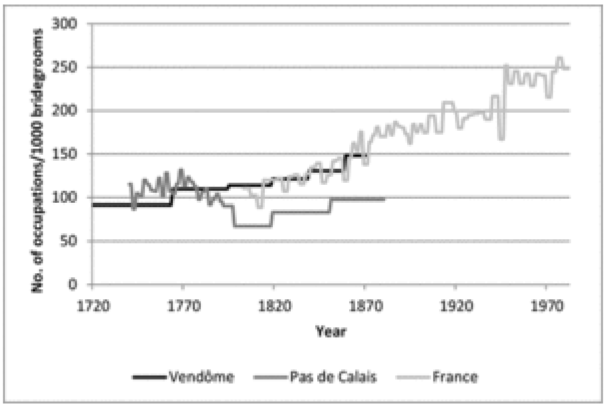

Figure 2. Occupational diversification in France, 1720–1986.

Source: van Leeuwen et al., 2016.

Industrialization has been defined by Davis (1955, p. 255) as "the use of mechanical contrivances and inanimate energy (fossil fuels and water power) to replace or augment human power in the extraction, processing, and distribution of natural resources or products derived therefrom." It is reflected in the labor force by the introduction of such mechanical contrivances as captured in occupational titles in two ways. The first relates to the rise of new types of work that could not have previously existed, such as that performed by telegraphers, machinery mechanics, or crane operators. The second is not reflected by entirely new occupational activities and titles but by the, usually gradual, introduction of mechanical contrivances into existing activities. Weaving, for instance, can be done by hand or by using a machine, and HISCO has different codes for a variety of weaving activities. These range from industrial ones—such as Beam Warper, Loom Threader (Machine), Cloth Weaver (Machine, except Jacquard Loom), Lace Weaver (Machine), Carpet Weaver (Machine), and Net Maker (Machine)—to manual ones, or to weavers and others about whose use or otherwise of machines one has no data. The first type can be defined as industrial, and the second and third types as non-industrial. This procedure underestimates the number of industrial workers in at least two ways. Some of the weavers for which the required information is unavailable might have used a machine, but they are included as non-industrial. Furthermore, some occupational activities, such as those of secretaries or bank employees, grew in number because of the growth in industrial activities, but they are not included here either. Figures 3 and 4 demonstrate the rise of industrial occupations, once again for a set of European countries (1800–1914) and for France (1720–1986).

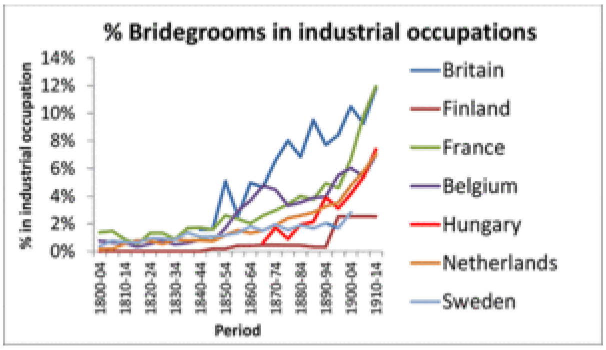

Figure 3. The rise of industrial occupations in Europe, 1800–1914.

Source: author's calculations on same datasets as reported in Maas & van Leeuwen, 2016.

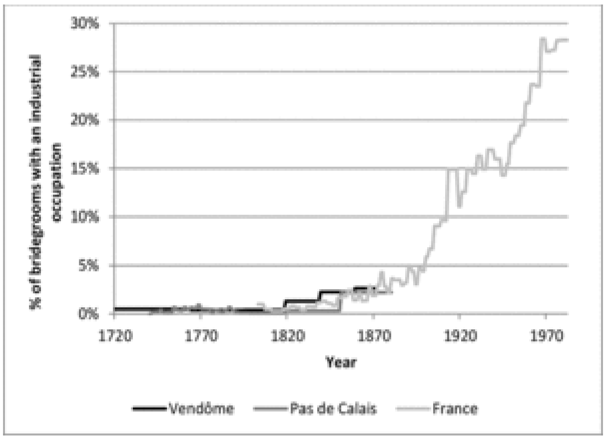

Figure 4. The rise of industrial occupations in France, 1720–1986.

Source: van Leeuwen et al., 2016.

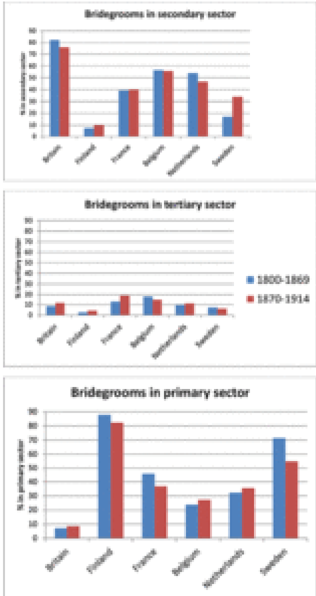

Figure 5. Changes in the size of the economic sector in Europe.

Source: author's calculations on same datasets as reported in Maas & van Leeuwen, 2016.

A related momentous change in the European economy over the past two centuries reflected in changes in the labor force has been the shift from primary (agriculture) to secondary (industry) to tertiary (service) sectors. For illustrative purposes, Figure 5 documents this for our set of European countries. The figures divide bridegrooms' occupations into the three broad economic sectors and two periods of time of equal length. A decline in the primary sector is visible, except where the primary sector had already greatly declined before 1800—in the United Kingdom, the Netherlands, and Belgium—and the consequent growth of the secondary and tertiary sectors, which generally also implies economic growth. There are, however, some caveats to this conclusion. These relate to the historical sources and not to the HISCO classification scheme per se, as all historical classification schemes based on vital registrations and censuses are affected. The first problem is that in many such historical sources there is a category "workers," "laborers," etc., and it is unknown whether they were agricultural or industrial workers. The solution to this problem taken in creating the HISCLASS social class scheme is to look at the occupational distribution of the communities these workers lived in: if the community is overwhelmingly rural, such persons are considered rural laborers; otherwise, they are considered industrial laborers. The second problem is that the occupations can be classified into economic sectors based only on the nature of the work, not on the sector in which individuals worked. A clerk is thus classified as working in the tertiary sector even if they worked in a steel factory. Historical vital registers are rarely so informative as to state in which company a person with a certain occupational title worked, while this might be known from industrial/labor force surveys. An additional issue worth mentioning is that up to the 21st century most coded historical occupations have been from Western countries. However, this issue has gradually been resolved. This opens the way for global historical comparisons (e.g., on economic specialization) at least for the male labor force —historical data with occupations of women are less readily available, though this too is changing.

Female Labor Force Participation

Large-scale digitization of censuses and vital registers has revealed that the historical record is more abundant in reporting women's work than previously thought (van Leeuwen & Zijdeman, 2014). As HISCO is almost gender neutral—though not completely, based as it is on ISCO68—there is no problem coding women's occupations. Source problems remain, of course, especially with regard to the varying degree of underregistration of certain types of women's work in censuses. Women (particularly married women) working in family enterprises and on farms have, in general, been particularly prone to undercounting. Such source problems are sometimes grave, depending on time, place, and source, and merit serious consideration that falls beyond the scope of this contribution. But there are sufficient datasets that shed light on male and female labor force participation and on the status of the work that men and women did. Before making some general remarks and showcasing some such recent studies, it is worthwhile stressing that, depending on time and place, historical vital records might undercount or overcount certain types of workers relative to a modern LFP measure. These sources simply indicate whether a person reported an occupation. This differs from the modern definition of working, or seeking work, within some specified period. Even with these sources, once can study occupational careers for women—having linked census data for men would make such a study slightly easier.

Over the past two centuries, two contrary processes operated. On the one hand, new jobs became available to women, first in industry, later in the service sector, increasing the range of work options for women (Charles & Grusky, 2004; Goldin, 1995). On the other hand, some types of work, such as the domestic production of textiles, suffered, and the male breadwinner/female homemaker household model became increasingly popular (Becker, 1981), leading to women retreating from the labor market until about the mid-20th century. This latter process was institutionalized by gender differences in education and sometimes legalized in mandatory retirement from work upon marriage. While no doubt more problematic than for men, female LFP can be documented using the HISCO family. Interestingly, Schulz, Maas, and van Leeuwen (2014) demonstrate that, initially, the average social status of women remaining in the labor force fell (due to the selective withdrawal of women from higher social groups); later, however, the status of working women rose again owing to the dissemination of gender-egalitarian values and a general rise in levels of education. Using commonly coded historical datasets, one can study "the quiet revolution" (Goldin, 2006) that increased women's horizons, identity, and decision-making. Female labor force participation today is generally both higher and more varied than during the mid-20th century and possibly earlier (see Zijdeman, van Leeuwen, Rébaudo, & Pélissier, 2014). Not only has it become possible to trace which women worked at any one time—and what caused geographical and temporal variations—it is also possible to study, using the same sources and methods discussed here, what female careers looked like. Schulz (2015) used a latent growth model to study what careers of Dutch single women look liked over the course of their lives, how this varied between places and changed over time, and what can explain these variations. She demonstrated that some modernization processes mattered: in municipalities with higher rates of mass communication, women's careers progressed more over the course of their lives. She also demonstrated that more women worked in municipalities with a top-100 company, presumably with internal procedures for hiring and promotion. Her multilevel research design thus demonstrated the importance of community-level institutions at both a concrete level (companies with internal labor markets) and a more diffuse one (gender values). In a broader sense, such studies enable us to test neo-institutional economic theories (North, 1990) on individuals in historical societies.

The studies just discussed draw attention to a few interesting observations. First, the LFP of women and men over parts of the globe over the past two centuries can be studied. Second, this can be done at any one point in time, by reference to the social origins of the persons involved or over the life course. Third, not only can these work processes be described, explanations can also be tested in a multilevel regression design with (admittedly a rather limited set of) historical explanatory variables at the level of the individual (microdata), the company, or community they lived in (mesodata), such as country- and time-specific phenomena. Examples of microdata include age and work experience (as human capital theory expects this to have a positive influence) and gender and migration status (since, for various reasons, including generally lower recognized educational qualifications, and discrimination by employers, women, and migrants might have less access to paid work, especially in higher-status jobs). Examples of mesodata at present relate to community characteristics such as the presence of companies with an internal labor market or gender norms as measured by means of equality in secondary schools or the degree to which the municipality is connected to the wider world through transport and communication (e.g., Lippényi, Maas, & van Leeuwen, 2015; Schulz, 2015; Schulz et al., 2014; van Leeuwen & Maas, 2019; Zijdeman, 2010). They might also relate to family characteristics in multilevel sibling models (Knigge, Maas, & van Leeuwen, 2014A; Knigge, Maas, van Leeuwen, & Mandemakers, 2014B). Country characteristics might include gender norms, evidenced for example in voting rights. In principle, the combination of micro-, meso-, and macro-historical data allow one to test economic and sociological theories using the immense geographical and temporal variations in LFP, types of jobs, and their status levels. There is still much to be done in terms of sourcing data for non-Western countries and additional data on women and of knowing how best to correct for underregistration, notably for ensuring a much richer set of both micro- and meso-explanatory variables. One approach is to use historical company records and connect these to data from vital registers and censuses. This is difficult due to the relative scarcity of such data, problems of record linkage, and a degree of incompatibility between functional descriptions of work in personnel files and occupational titles in vital registers and censuses. But it would enable us to historicize what, in retrospect, seems to have been the golden age of steady employment and upward occupational mobility, at least in the West. In retrospect, this was a rather brief period, sandwiched between eras dominated by temporary jobs and meandering careers (Brown, van Leeuwen, & Mitch, 2004). One would also be in a better position to test theoretical explanations of these changes.

The changes in the labor market observable with simple measures based on a comparative historical occupational coding scheme such as HISCO and discussed so far have a bearing on economic processes. But occupational titles are also indicative of social position, and by systematically recoding HISCO into historical measures of class and status, one can study not only important transformations in the economy over the past centuries but also concomitant huge transformations in the global social fabric.

HISCO-Based Measures of Class and Status

Cut down a tree and observe its rings. Or take an early walk on the beach and disturb the tidal deposits of sand running in more or less parallel lines. Or, and this is where the image of strata comes from, look at the wall of a quarry or a canyon and discern the various layers, of different hues and widths, which, if you are lucky, run parallel to one another, from the base to the peak. Transferring this image to human societies, these deposits can be said to capture strata. Human societies are stratified. Their inhabitants possess valuable goods, power, prestige, and other scarce desirables in unequal amounts. On this there is much consensus among social scientists, but not on some related matters (van Leeuwen, 2020). Are there many layers, or just a few? Do they run perfectly in parallel to one another—from low to high status—or at sharp angles, with perhaps fractures and crossovers? Are they the same for every society on the globe, and each time period, or do they vary markedly? And how does one construct such measures?

Without consistent occupational measures, comparisons in space and time are impossible or flawed. The makers of HISCO were certainly not the first to acknowledge this, nor are they the first to tackle this problem. In fact, they stood in a tradition of generations of occupational statisticians, stratification sociologists, and social and economic historians. The basic building block of the economic and social fabric of work as displayed in occupational titles—HISCO—is a historical version of an international scheme developed by the ILO and used around the globe by statistical agencies. While it allows for comparison and reduces complexity, its 1,675 categories are too many for many purposes. For questions on social structure and the effects of social inequalities on lifestyles and life chances, stratification sociologists and economic and social historians usually work either with a class scheme or with a status scale. For comparative studies on work, economists, historians, demographers, and many others need comparative measures that can, where possible, bridge geographical regions and historical eras. Standing in this research tradition, in a collaborative effort, a measure of social status (HISCAM) and a measure of social class (HISCLASS) were constructed.

HISCAM: A Continuous Historical Status Scale

To create a continuous historical status scale based on occupations, one cannot use the well-established procedures involved in constructing prestige or socio-economic status scales. In both cases data are needed from individual respondents that are found only rarely in historical documents: their stated opinion on the degree of prestige commonly associated with an occupation or the amount of income and education typically associated with it. Lacking such data, one may turn to another type of social status scale (interaction scale) that can be constructed from occupational information in vital registers and occupations once these have been coded into HISCO (Lambert & Griffiths, 2018).

For this purpose a HISCO-based scale, HISCAM, has been developed (Lambert, Zijdeman, van Leeuwen, Maas, & Prandy, 2013). The scale was derived using patterns of occupational connection. The idea behind this scale is that occupational categories that interact frequently (e.g., there being many marriages between people having these occupations) are similar with respect to social status. Such an interactional approach to social status has the advantage that it can be derived from occupational data commonly available for historical populations. The frequency of occurrence of all particular combinations, or interactions, of occupations is modeled (e.g., how many bakers married daughters of bakers, but also how many bakers marry butchers, secretaries, mayors, etc.). Without going into the technical details, a score is assigned to each occupation to indicate its position within the empirical dimension of social interaction, with higher scores indicating a more advantaged position in society. This analysis was performed on data for the period 1800–1938, principally derived from marriage registers, encompassing over 2 million intergenerational relationships. Using this procedure, a single "universal" scale has been estimated for the full span of data as well as scales for a particular country or period. HISCAM is a continuous scale, and it can be used in regression analysis to explain and to be explained.

HISCAM will enable researchers to scrutinize the claim of invariability of social status across time and space, the so-called "Treiman constant." According to Hout and DiPrete (2006), researchers of social stratification—and even sociologists in general—have discovered only one universal: occupations are ranked in the same order in most nations and over time. Because this conclusion was based mainly on a comparative study by Treiman (1977), it is termed the Treiman constant. According to classic functionalist reasoning, some occupations have higher prestige and income because they require special talents or training or they are functionally more important. The rare talents and great efforts needed to perform important occupations well need to be rewarded by "society"; this is achieved through prestige and remuneration. Treiman's theory of prestige extends this reasoning to international and historical comparisons. Because all societies face similar functional imperatives, a similar division of labor develops. The more powerful and privileged occupations are the same in all societies, and they are always more highly regarded.

Treiman found support for his claim in the high correlations between the prestige rankings of occupations in different countries and different periods. Most of the rankings in his study are from the second half of the 20th century, but he also presents similarities with a Nepalese ranking of casts from 1395 and an ordering of guilds in Florence in 1427. Other studies, comparing 19th-century with 20th-century rankings, support the Treiman constant as well. Notwithstanding his general conclusion, Treiman noted that certain occupations deviated from the general pattern. Manual workers had higher prestige in socialist countries, and office clerks lost prestige when literacy in society spread, just as religious officials might in this day and age of secularization see their status diminish somewhat. There is some historical and regional specificity for some occupations in 19th-century Europe. A preliminary conclusion is that Treiman's claim is largely supported for contemporary societies, but that it leaves room for historical deviations in the case of specific occupations. Scholars are in a position to document and explain the prevalence, size, and causes of such deviations.

HISCLASS: A Historical International Social Class Scheme

Sometimes scholars do not want a continuous ranking of occupations in terms of social standing into maybe a thousand steps ranging from 0 to 100; instead, for theoretical or practical reasons, they require grouping into just a dozen or so "large" social classes. These classes should follow major fault lines in the social fabric of contemporary or historical societies. Social stratification and the division of society into a limited number of large classes are themes that have been extensively discussed in the social sciences ever since the origin. However, it was only after World War II that methodological strategies were developed operationalizing concepts of social class. Following the traditions established by Marx and Weber, various researchers developed social class schemes based mainly on occupational titles. It can be said that a social class is a group of people with the same opportunities in life. HISCLASS has the HISCO-based historical social class scheme as its main dimensions: the division between manual and non-manual occupational skills, the extent to which some supervise the work of others, and the economic sector. In this way, the HISCLASS scheme seeks to conform to the way economic and social historians have looked at social stratification in historical perspective. Table 2 shows how the 12 HISCLASS categories are derived from the main class dimensions (van Leeuwen & Maas, 2011).

Table 2. HISCLASS: 12 Classes Based on 4 Class Dimensions

Class No.	Class Label	Manual/Non-Manual	Skill Level	Supervision	Sector
1	Higher managers	Non-manual	High	Yes	Mainly Other
2	Higher professionals	Non-manual	High	No	Other
3	Lower managers	Non-manual	Medium	Yes	Mainly Other
4	Lower professionals and clerical and sales personnel	Non-manual	Medium	No	Other
5	Lower clerical and sales personnel	Non-manual	Low	No	Other
6	Foremen	Manual	Medium	Yes	Other
7	Medium-skilled workers	Manual	Medium	No	Other

8	Farmers and fishermen	Manual	Medium	No	Primary
9	Lower-skilled workers	Manual	Low	No	Other
10	Lower-skilled farm workers	Manual	Low	No	Primary
11	Unskilled workers	Manual	Unskilled	No	Other
12	Unskilled farm workers	Manual	Unskilled	No	Primary
Source: van Leeuwen and Maas (2011).					

In order to allocate systematically the 1,675 HISCO occupational groups to their appropriate classes, one could safely rely primarily on the judgment of expert historians, who generally do a good job. However, to create HISCLASS, a more quantifiable method was used, processing information found in the *Dictionary of Occupational Titles* (DOT), compiled by the United States Employment Service. Ever since the 1930s, North American occupational analysts have gone to farms, shops, factories, businesses, and other places of work to observe men and women at work. They collected information on tasks performed, knowledge required, machine equipment and materials used, physical demands and working conditions, and required worker characteristics. The first published DOT dates from 1939, and since then there have been many updates, including an Internet incarnation OHNET. For HISCLASS, a concordance was made between HISCO and the DOT (1965). To generate HISCLASS based on HISCO and the DOT, first each of the 1,675 categories in HISCO was matched to one of the 10,000 categories found in the DOT. The numeric information (e.g., on skill level) in DOT could thus be attributed to each HISCO category. Subsequently, this information was used to score the HISCO category on each of the HISCLASS dimensions and thus to place it in the appropriate social class. This operation was then scrutinized by expert historians to addresses the potential problem of applying information from a mid-20th-century occupational dictionary to earlier periods. Whenever the majority of the historians consulted disagreed with an original proposal, the classification criteria were re-evaluated. In this way, HISCLASS has the advantage of being both systematic and having been extensively tested by historians. There is a recode job from HISCO to HISCLASS on the HISCO website.

Having a historical class scheme and the necessary historical data, one can study comparatively and quantitatively the evolution of class structures over time. Figure 6 shows this for men in France for 1803–1986, for which period there are national data as opposed to earlier local or regional data. The 12 hisclasses have been amalgamated into 5 classes. With a little more detail one can see what has already been noted. Over the past two centuries, the rural classes of farmers and farmworkers shrank in number in comparison with those of skilled workers and with non-manual workers in the service class. A similar transition took place in other European countries. However, as yet there has been no comparative series on changing class distributions in various parts of the globe over the past two or more centuries. Nor has the use of occupational characteristics (and how these evolved) based on a combination of HISCO and the many DOT characteristics been exhausted. One thing that has been picked up (e.g., de Pleijt, Nuvolari, & Weisdorf, 2019) is that HISCLASS classifies occupations into several skill levels, skill being understood as both specific vocational training as well as its required level of general education. Once occupations are coded into HISCLASS, one can recode them into skill levels to study, for instance, the effect of the various industrial revolutions on the evolution over time of the distribution of the labor force according to skill level. In a way, for historical populations this supplements the absence or presence of a signature in a vital document as an indicator of educational achievement, albeit only with regard to the level of education typically needed for an occupation. To go further, one would need to combine such sources as historical vital registers with school data or with time- and place-specific information such as the educational notes on their congregations taken each year by Swedish ministers of religion. This would enable us to better test explanations of long-run economic growth that emphasize human capital (Galor, 2011; Mokyr, 2017).

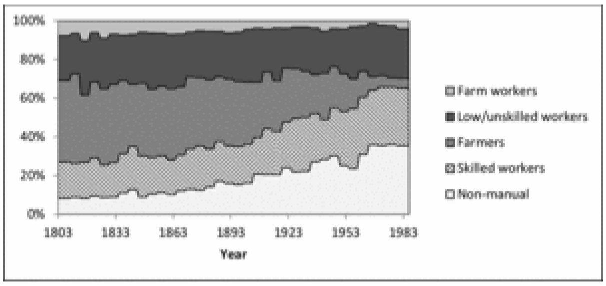

Figure 6. Class distributions of grooms in France, 1803–1986.

Source: van Leeuwen et al., 2016.

If the class structure changes, so a large body of the sociological literature reasons, this must have an effect on intergenerational social mobility. If the social class of your parents were to become extinct, you would have to earn your livelihood in another social class, leading to an increase in intergenerational social mobility. Next to the “direct” consequences of such changes in the labor force, modernization theorists have long argued that there have also been other influences of social openness. In a notionally static society of farmers and shopkeepers, farms and shops are usually passed down the generations: the required capital, skills, and social connections are either inherited or acquired through socialization. But once the nuclear power plants open, for instance, new occupations come into being that parents cannot transfer to children. The past two centuries have witnessed the growth of such new specialized occupations, and for this reason modernization theorists expect growth in class mobility. They expect this too because of concomitant processes such as educational expansion—where children and young adults learn at schools skills not possessed by their parents. These skills open up new jobs, where they might meet people from other social classes, and where they might be imbued with universalistic values that tend to diminish class differences and to stress everyone’s freedom to choose their own occupational path. Modernization theorists such as Treiman (1970) and Landes (1969) also argue that as societies “modernized,” in order to maximize efficiency in modern workplaces, industrial employers increasingly recruited their personnel based on individual merit rather than on the merits of their parents, leading to a loosening of direct bonds between class of origin and destination, a transition from ascription to achievement.

However, other scholars, such as Bourdieu and Erikson and Goldthorpe, have doubted this rather optimistic picture and stressed the ability of elites to circumvent, for instance, the leveling effect of education by sending their children to prestigious schools, by helping them with their homework, by paying for extra tuition, etc. After a very long debate, the general consensus is that during the 20th century male social mobility increased from one generation to the next (Breen & Luijkx, 2004; Ganzeboom, Luijkx, & Treiman, 1989).

Figure 7 shows the growth of observed intergenerational social mobility in France over more than two centuries (van Leeuwen et al., 2016). The data from the most recent period come from a national sample of marriage registers; for the other two periods they use the same source but relate to one province and one provincial town, respectively. Upward and downward mobility are counted in the same way. Figure 8 shows the estimated growth in relative mobility—net changes in the distribution of the labor force in France over the same period. The reader should be aware that the figure follows the convention in stratification research in using a measure of relative closedness. This means that if the trends go downward, relative mobility, or openness, increase.

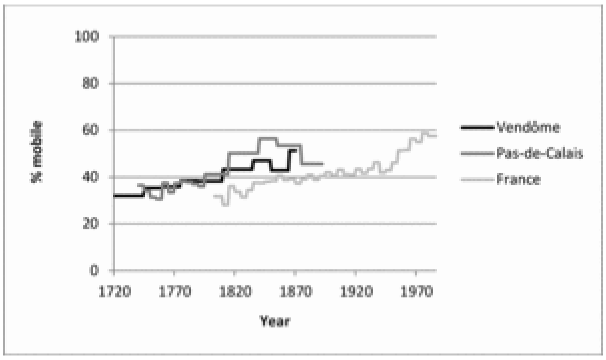

Figure 7. Total intergenerational mobility in France, 1720–1986.

Source: van Leeuwen et al., 2016.

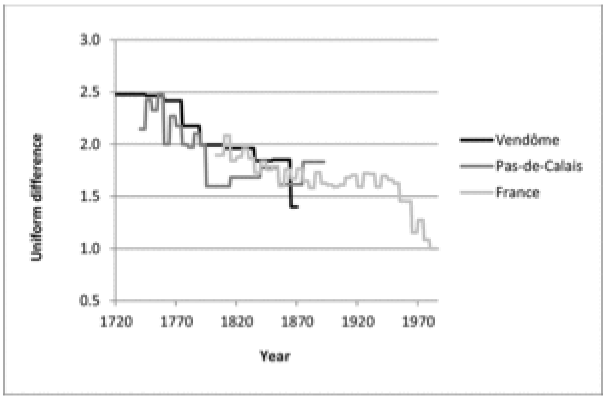

Figure 8. Relative mobility in France, 1720–1986.

Source: van Leeuwen et al., 2016.

Note: General association according to a uniform difference model with multiplicative parameters indicating how much greater or smaller the association in a certain mobility table is compared with that for France in 1980–1986.

As is also the case for the European countries in our sample during the shorter period for which there are comparative data, observed, total, social mobility, and estimated relative mobility increased over time. So modernization theorists seem to have a point, although these simple figures cannot of course determine whether the causes of this change are those assumed by the theory; for this, a multilevel research design is needed. What is known is that both total and relative social mobility were higher in more industrialized countries than in less industrialized ones and increased in most countries after industrialization.

A Historical Microclass Scheme

A social status scale like HISCAM measures social status continuously. Society is thus neatly and finely ordered. Large social class schemes such as HISCLASS have less detail—they might have only a dozen or so classes—but also less order, in the sense that not all classes can be ordered perfectly from high to low. Especially in the middle of such a class distribution, things get fuzzy and it is not clear which class is higher than the other. Large class proponents will say this is how it ought to be, as it reflects fuzziness in real social stratifications.

In between continuous status scales and large class schemes lies the novel theoretical concept of microclasses. Weeden and Grusky (2005) have argued that the large classes hide too much variation, relevant for example to explain the transmission of inequalities between the generations. To uncover this variation, it is argued that microclasses are needed. These can be seen as occupational networks through which resources (human, cultural, and social capital) are exchanged, creating layers of stratification both within and between the large classes. For instance, office and clerical workers are usually grouped within a single microclass, as they are likely to have broadly similar educational backgrounds and broadly similar knowledge of how the workplaces operates. Bakers and butchers, however, are usually given their own separate microclasses as the specialist skills and knowledge required for their roles cannot be gathered from other occupational positions. This is to say office workers possess knowledge of the processes, norms, opportunities, and expectations of a wide variety of office jobs, whereas bakers have specialist knowledge which cannot be translated to any other forms of work. Since microclasses are more fine-grained—a typical number is a hundred—and may be found within and between large classes, they enhance our understanding of occupational stratification and its consequences for inequality outcomes studied in economics, sociology, and the life sciences.

A recent historical comparison of intergenerational male social mobility patterns in 64 microclasses during early industrialization did indeed demonstrate that there has been a substantial degree of social mobility within large classes that can be observed only through a microclass approach. In historical societies, people experienced intergenerational social mobility between microclasses. That did not always translate into a changing position between large classes or into a changing relative position in a historical status scale (Griffiths, Lambert, Zijdeman, van Leeuwen, & Maas, 2019).

Measures and Procedures to Study Long-Term Changes in Economy and Society

Just a few decades ago economic historians had neither the data nor the methods to study comparatively the work of women and men across various parts of the globe over past centuries. In the second quarter of the 20th century they have. This is due in part to the large-scale digitization of historical records, in part to new research efforts to find such data for women in the West and for men and women in other parts of the world, and in part due to new methods and procedures. Some of the latter have been discussed here. The comparative global coding scheme HISCO allows us to study processes of economic diversification and labor force participation by means of a set of procedures. These procedures—which include looking at occupational diversification and specialization—are part of the HISCO family, which also includes a continuous status scale (HISCAM), a large social class scale (HISCLASS), and a HISCO-based microclass scheme. While one has to acknowledge some lacunae in data and knowledge, and this article has suggested how these might be resolved, taken together the wealth of historical data and the availability of tools from the HISCO family allows economic historians to comparatively document and explain the varying historical trajectories of the wealth of nations, the prosperity of communities, and the opportunities for individuals to find meaningful, well-paid, and steady work, with a measure of freedom from their social origins and a measure of social and occupational mobility over the life course.

In a sense, constructing this family of HISCO tools, and making good use of them, never ends. There is room for improvement—by including more studies on women (some were discussed in the section on female labor force participation) and on non-Western societies (although HISCO has now been applied in every continent). It is also true that there have been fewer studies on careers in relation to intergenerational social mobility and social homogamy. There is also room to expand HISCO-based studies on the historical development of the labor force.

The past two centuries have witnessed major technological innovations in industry, services, and agriculture, including the mechanization of spinning and weaving, the rise of factories, the use of steam power and of electricity as a source of energy, chemical innovations, and the digital revolution. These innovations might affect the demand for certain types of labor, and differentially influence wages and employment rates. Machines and routines might at times replace certain types of skilled worker, such as master artisans, leading to unemployment or job degradation and lower wages. Or it might in other places and times lead to increasing demand for certain highly skilled occupations needed to operate the new, complex, and costly machines and lead to an increase in the skill level, and occupational social status, of the labor force at large or parts of it. Historical economists distinguish between skill-biased technological change (SBTC) and routine-biased technological change (RBTC) (Acemoglu & Autor, 2012; Autor, Levy, & Murname, 2003; Goldin & Katz, 2008). SBTC enhances the skill level of the labor force overall; RBTC does so only for routine-based work and affects people working in such occupations, but not those in non–routine-based occupations, either at the higher or the lower end of the skill distribution. Which process is at work in a particular society is a question for empirical inquiry.

Historical census, or workplace, data on occupations and wages can be of use to describe such differential changes across occupations, economic sectors, time periods, and places. The occupational data need not per se be coded in (H)ISCO. To study a limited region or time period, any reasonable local or national coding scheme will do. But to document changes over time and to compare across regions, consistent coding is essential. As many censuses and labor surveys are coded in versions of ISCO anyway, using HISCO makes it relatively easy to connect to these. Coding into HISCO might also be advantageous in that good use can then be made of the crosswalk between HISCO and the *Dictionary of Occupational Titles*, as it connects occupations to skill levels but also to a variety of other traits. Using HISCLASS, one can begin to study long-term changes in the distribution of skill levels comparatively. Using the concordance underlying HISCLASS, one might further study temporal and spatial variation in a variety of cognitive, interpersonal, motoric, and physical skills, although ideally this would demand a series of crosswalks between HISCO and DOT.

Links to Digital Materials

EHPS-Net, European Historical Population Samples Network.

HISCAM.

HISCO.

History of Work Information System.

INCHOS, International Network for the Comparative History of Occupational Structure.

IPUMS, Integrated Public Use Microdata Series.

IPUMS-I, Integrated Public Use Microdata Series, International.

MOSAIC project on Recovering Surviving Census Records.

O*NET.

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